

Balance equation

$$k_c \frac{d^2 T}{dx^2} + \dot{q} = 0$$

$$\frac{d^2 T}{dx^2} = -\frac{\dot{q}}{k_c}$$

integrate twice

$$T_f(x) = -\frac{\dot{q}}{2k_c} x^2 + C_1 x + C_2$$

domain $L \leq x \leq L+b$

BCs

$$\left. \frac{dT_f}{dx} \right|_{x=-L} = 0$$

$$T_f(L) = T_s(L)$$

$$k_f \left. \frac{dT_f}{dx} \right|_{x=L} = k_s \frac{dT_s}{dx}$$

$$-k_s \frac{dT_s}{dx} = h[T_s(L+b) - T_\infty]$$

$$C_1 = -\frac{L\dot{q}}{k_F}, \quad C_2 = \frac{3L^2}{2hk_Fk_S} \frac{hk_S\dot{q} + 4Lbhk_F\dot{q} + 4Lk_Fk_S\dot{q}}{2hk_Fk_S}$$

$$A = -\frac{2L\dot{q}}{k_S} \quad \beta = \frac{2L^2h\dot{q} + 2Lbh\dot{q} + 2Lk_S\dot{q} + T_{\infty}hk_S}{hk_S}$$

$$T_f(x) = T_{\infty} + \dot{q} \left(\frac{3L^2}{2k_F} + \frac{2Lb}{k_S} + \frac{2L}{h} - \frac{L}{k_F}x - \frac{x^2}{2k_F} \right)$$

$$T_s(x) = T_{\infty} + \dot{q} \left(\frac{2L}{h} + \frac{2L^2}{k_S} + \frac{2Lb}{hL} - \frac{2L}{k_S}x \right)$$

